

Biomarker correlates of reaction time with false-discovery-rate control

Abstract

Background: A laboratory study of caffeine and attention. Methods: Each of the six biomarkers was correlated with reaction time (ms) on the Stroop task using Pearson correlation (two-sided). Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a laboratory study of caffeine and attention. The primary question concerns caffeine administration and reaction time (ms) on the Stroop task. The accompanying dataset (data.csv) contains one row per participant-record as exported from the study database, including some duplicated exports and sentinel-coded missing values; it is described in the data dictionary.

Methods

Participants and data. The export contains 390 records. Variables are listed in the data dictionary. Group membership is recorded in `condition`.

Data cleaning and exclusions. Exact duplicate records (identical on every column) were removed, keeping the first occurrence. The value -999 denotes a missing measurement and was treated as missing. Only participants aged 18 to 80 years inclusive were analysed. Records with missing `rt_ms`, `biomarker_1`, `biomarker_2`, `biomarker_3`, `biomarker_4`, `biomarker_5`, `biomarker_6` were excluded (complete-case analysis).

Statistical analysis. Each of the six biomarkers was correlated with reaction time (ms) on the Stroop task using Pearson correlation (two-sided). The six p-values were adjusted for multiple testing with the Benjamini-Hochberg false-discovery-rate procedure; associations with adjusted $p < 0.05$ were declared significant. The strongest association is the biomarker with the largest absolute correlation. All analyses used the cleaned analysis sample; no random resampling was used.

Results

Among 344 participants, `biomarker_1` correlated with the outcome ($r = 0.455$, adjusted $p < 0.001$), as did `biomarker_3` ($r = -0.325$, adjusted $p < 0.001$); the adjusted p for `biomarker_5` was $= 0.094$. In total 2 biomarkers remained significant after adjustment; the strongest association had $r = 0.455$.

Discussion

The analysis followed the pre-specified plan. Limitations include reliance on database exports and complete-case handling of missing values.